Computer Networks Spring 2026
Assignment#5 (6A & 6B)

Due Date: Thursday, April 23rd, 2026

Submission Mode & Time: Handwritten solutions to be submitted during the lecture.

Please note the following:

1. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
2. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
3. All solutions must be **hand-written.**
4. **Assignment Solution Submission**: In case of **in person / physical lectures at the campus**, hard copy of the hand-written assignment's solutions will be submitted by **hand** by each student to the Instructor / TA directly during the lecture on the due date.

PART-1

Use the following text for completion of this part of the assignment:
Computer Networking - A Top-Down Approach 8th Edition by Kurose & Ross.

Solve the following problems from the back of **Chapter 5**. Every Question has equal marks i.e.

Problems: **(4*5 = 20 marks) [CLO 3]**

P14, P15, P17, P18, P19

PART - 2

Question: 01 [4 Marks] [CLO3]

Consider an AS with routers A, B, C, D & E with the topology A—B—C—D—E where hop-count is the metric. A goes down. B does not receive the vector update from A so it concludes that its route of cost 1 to A is no longer available. C doesn't know yet that A is down and tells B that A is 2 hops away from it. Thus, the wrong info propagates until it reaches infinity. Briefly give the solution to this problem?

Question: 02 [6 Marks] [CLO 3]

Suppose Autonomous Systems, i.e. AS X & AS Z are not directly connected but are instead connected by AS Y. Further suppose that X has a peering agreement with Y, and that Y has a peering agreement with Z. Finally, suppose that Z wants to transit all of Y's traffic but does not want to transit X's traffic.

a) Does BGP allow Z to implement this policy? (Yes or No)

b) If No, why not? If Yes, how? (Give at least two reasons for either case.)

Question: 03 [4 Marks] [CLO 3]

1. How does BGP use the NEXT-HOP attribute? How does it use the AS-PATH attribute?
2. Describe how a network administrator of an upper-tier ISP can implement policy when configuring BGP.

Question: 04 [6 Marks] [CLO 3]

Consider the following AS relationships:

- AS1 is a provider for AS2 and AS3.
- AS2 and AS3 are Peers.
- AS2 is a provider for AS4.
- AS3 is a provider for AS5.

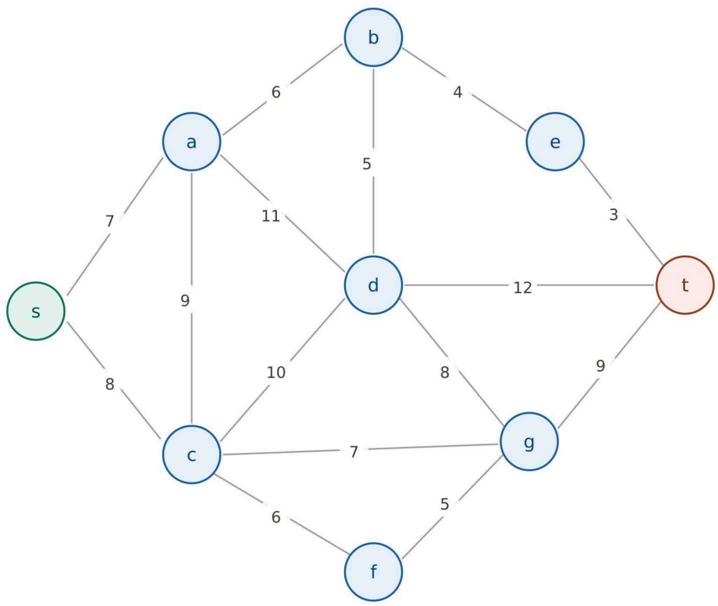
AS2 learns a route to a destination inside AS4.

1. Will AS2 advertise this route to AS1? Why or why not?
2. Will AS2 advertise this route to AS3? Why or why not?
3. AS2 learns a route from AS3 (its peer). Will AS2 advertise this route to AS1?

Question: 05 [12+2+3 = 17 Marks] [CLO 3]

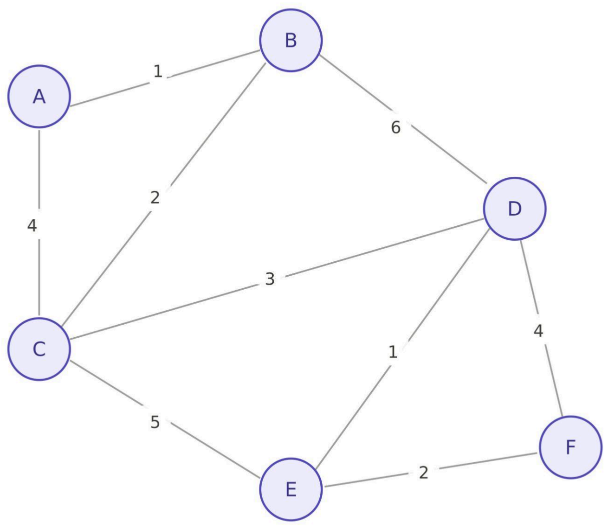
Consider the undirected weighted network below with nodes {s, a, b, c, d, e, f, g, t}.

a. Using Dijkstra's shortest-path algorithm, compute the shortest path from s to all other nodes in the network. Show your work by completing a table similar to Table 5.1 of Computer Networking - A Top-Down Approach 8th Edition by Kurose & Ross.
b. List the shortest path from s to t and give its total cost.
c. Suppose the cost of edge (c, f) is reduced from 6 to 2. Determine whether the shortest path from s to t changes, and justify your answer.



Question: 06 [6+12+6+4 = 28 Marks] [CLO 3]

A small enterprise network has six routers {A, B, C, D, E, F } connected by bidirectional (undirected) links. The link costs represent administrative distance / routing metric. Each router uses the Distance-Vector (DV) routing protocol: it maintains a distance vector to every destination and periodically shares it only with its directly connected neighbors.



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a. Write down the initial distance vector for each router (before any exchanges). Each router knows only its directly connected link costs; all other entries are ∞.
b. Simulate three rounds of synchronous DV updates. In each round, every router sends its current distance vector to all direct neighbors, and every router updates its own vector using the Bellman-Ford

equation:

$$D_x(y) = \min_{v \in \text{neighbors}(x)} \{c(x, v) + D_v(y)\}.$$

Show the distance vector of every router after Round 1, Round 2, and Round 3, using the table format below:

Round	D(·, A)	D(·, B)	D(·, C)	D(·, D)	D(·, E)	D(·, F)
0 (init)						
1						
2						
3						

c. I. Has the DV protocol converged after Round 3? Justify by checking whether any entry changes in a hypothetical Round 4.
II. State the shortest path (sequence of routers) and cost from A to F . Verify it matches the Bellman-Ford optimal.
III. How many rounds does convergence require in the worst case for a network with diameter d (longest shortest path in hops)? Relate this to the structure of this network.

d. Suppose that after convergence, the cost of link (D, E) increases from 1 to 10 (simulating link degradation, e.g., high error rate). Describe step-by-step how the count-to-infinity problem manifests between routers D and E for the destination F . Trace at least four DV-exchange rounds showing the erroneous distance estimates.